

Project Proposal: The Post-Purchase Concierge (PPC)

An Autonomous Agentic System for Warranty Tracking and Return Optimization

1. Problem Statement & Motivation

Real-World Business Problem

In the modern retail landscape, consumers face significant "return friction" due to inconsistent return windows (ranging from 14 to 90 days) and obscured warranty terms. For a graduate student and tech enthusiast, high-value purchases like a **Samsung 65" Neo QLED TV** (\$1,155.99) or a **Canon EOS R50 Camera** (\$799.99) represent significant investments. Missing a 14-day return window due to administrative oversight results in "financial leakage"—where users keep sub-optimal products simply because the "cost" of tracking the deadline was too high.

Target Users

The primary users are high-frequency electronics consumers and "power shoppers" who manage multiple high-value transactions across various retailers. These users value their time and require a high level of accuracy to protect their financial investments.

Why an Agentic AI Solution?

Traditional notification apps fail because they require manual data entry, which is the primary point of failure. An agentic solution is appropriate here because it provides **autonomous reasoning** and **proactive execution**. Unlike a human-defined workflow, the agent must:

- **Analyze Unstructured Data:** Read messy, semi-structured email receipts to extract product names and dates.

- **Autonomous Research:** Dynamically search the web for a retailer's specific policy, which may change seasonally.
 - **Cross-Platform Integration:** Coordinate between an email inbox (Gmail), a scheduling tool (Google Calendar), and a notification system (Apple Watch/Mobile).
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2. Use Cases & Scenarios

Scenario A: The "Open Box" Anomaly

- **User Scenario:** On January 30, the user purchased an **Open-Box Samsung TV**. Open-box items often have stricter return rules than new items.
- **Agent Tasks:** **The Inbox Scout** identifies the receipt; **The Policy Researcher** confirms the 14-day limit for open-box electronics at Best Buy; and **The Scheduler** blocks a "Final Inspection" date on the user's calendar for February 11.

Scenario B: Post-Delivery Quality Audit

- **User Scenario:** The user receives a **Canon R50 camera** but is preoccupied with school coursework.
 - **Agent Tasks:** **The Extraction Agent** identifies the delivery date via the Gmail API; **The Planner** calculates the window; and **The Notifier** sends a haptic alert to the user's **Apple Watch** on day 12, asking: "Have you tested the autofocus? Your return window closes in 48 hours".
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3. Initial System Design

Preliminary List of Agents and Roles

- **The Inbox Scout (Data Fetcher):** Monitors the Gmail API for keywords like "Best Buy" or "Order Confirmed".
- **The Extraction Agent (Summarizer):** Uses GPT-4o to parse text and extract the Product Name, Price, Date, and Store.
- **The Policy Researcher (Web Search):** Uses the Tavily or DuckDuckGo API to find current 2026 retailer return policies.
- **The Scheduler (Planner):** Interfaces with Google Calendar to autonomously block "Decision Deadlines".
- **The Notifier (Critic/Dispatcher):** Verifies the date accuracy and sends haptic alerts to the user's Apple Watch via the Telegram Bot API.

Expected Tools/APIs & Data Sources

- **APIs:** Gmail API, Google Calendar API, OpenAI API (Reasoning), Tavily (Search).
 - **Execution:** Python-based environment utilizing Pandas for local data logging.
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4. Feasibility & Risks

Potential Challenges

- **Parsing Logic (Hallucinations):** Emails vary by retailer. **Mitigation:** Using LLM-based parsing (GPT-4o) rather than rigid code to handle messy receipt text.

- **Tool Failures:** If the search API returns a general policy instead of the specific "2026 Holiday Policy." **Mitigation:** The agent will cite its source URL for user verification.

Constraints

- **Privacy:** Handling financial data requires local handling of API credentials.
- **Rate Limits:** "Student Tier" API limits will be managed by scanning only the last 25 relevant emails.

Timeline for Project Execution

- **Weeks 1-4:** API Foundation (Gmail/Calendar) and Metadata Extraction.
- **Weeks 5-8:** Web Search Integration and Policy Summarization.
- **Weeks 9-11:** Notification Bridge (Telegram/Apple Watch) and Final Demo Prep.
- **Deadline:** April 27, 2026.